

The Topological Architecture of Computational Transformations

Positional Continuation, the Kulik Gap Operator, and the Non-Linear Instantiation of Data Spaces

Driven by Dean A. Kulik

August 26

Introduction: The Transformation Substrate and the Instantiation of Data

In classical information theory and computational modeling, a discrete string of data—such as a 64-character hexadecimal SHA-256 digest or a compiled binary executable—is conventionally treated as a static repository of payload. Under standard Kolmogorov complexity frameworks, the sequence is analyzed strictly as a compressed or uncompressed representation of a specific state within a finite phase space. However, emerging paradigms in discrete topology, causal structuring, and algorithmic information theory dictate a fundamental inversion of this model: discrete strings do not passively store data waiting to be read. Instead, they function as foundational topological substrates that generate mathematically exhaustive addressable state spaces when subjected to continuous, lawful transformation.

The analysis presented herein explores this architecture by examining the structural mechanics of positional continuation, geometric admissibility, and topological severance. At the core of this framework is the recognition that computational spaces are fundamentally mathematical spaces defined by *shape*. Transformations that chart distinct trajectories over time constitute measurable, operational shapes. Crucially, when examining transformations across differing bases—such as the conversion of a structured hexadecimal sequence into a decimal format—the translation is emphatically non-linear. A consistent, stacked, or linear shape within the source hex does not yield a 1:1 consistent pattern in the decimal output. This non-linear mapping reveals the hidden informational capacity embedded within the transformation itself.

To fully conceptualize this mechanism, it is necessary to treat compiled hex not merely as flat data, but as a dense field of potential containing hidden structural gaps. Analogous to a compiled executable program, the raw hex on a disk does not visibly contain the dynamic states of the application—whether those states represent complex DNA sequence calculations or a fully rendered interactive game. The static hex consists

Copyright Dean A. Kulik – Orcid ID # 0009-0003-3128-8828

Creative Commons Attribution-NonCommercial 4.0 International License (CC BY-NC 4.0)

github.com/QuHarmonics/The-Nexus-Harmonic-Reality info@quharmonics.com

of indistinguishable continuity. The dynamic application only emerges when a reader (such as a CPU's instruction pointer) navigates the hex, introducing gaps, steps, and waits for input. The output is not added to the file; it is instantiated by the specific boundary conditions and operational masks applied to the continuous substrate.

This framework formally anchors the mechanics of data instantiation in verified structural mathematics, demonstrating why a single 256-bit hash can hold approximately 290 exabytes of distinct, addressable outputs without violating the conservation of mass or memory. The system is not storing the data; it is storing a singular continuous sequence, H , while an operator tree dictates exactly how history folds through that sequence to construct the future.

The Mechanics of Path Preservation: Positional Continuation vs. Arithmetic Collapse

The distinction between standard arithmetic accumulation and positional continuation serves as the primary gateway to understanding how discrete strings generate structural volume. In traditional arithmetic, operations inherently destroy the historical path of their operands.

Consider the commutative equivalence $2 + 3 = 3 + 2 = 5$. The resulting scalar projection retains no memory of the specific sequence of operations that produced it. Arithmetic destroys the path because it treats both operands as having equal historical weight. The geometric origin of the components is permanently erased by the operation, resulting in an absolute loss of structural history.

Conversely, positional continuation preserves the historical path by ensuring that the first event is structurally amplified by the mere existence of the second. Given a base B , the positional transition is expressed as an ordered polynomial recursion. The operation $2B + 3$ yields a fundamentally different physical state than $3B + 2$. For base 16, $2(16) + 3 = 35$, whereas $3(16) + 2 = 50$. The same distinct values, subjected to a different path, yield different resultant states because the continuation operation rescales the history of the system.

This absolute structural truth is defined by the principle: *the future gives the past its weight*.

The mathematical engine driving this preservation is isomorphic to polynomial evaluation via Horner's method. Horner's scheme optimizes polynomial evaluation by rewriting expressions as a nested sequence of multiplicative and additive operations. For a discrete sequence of digits d_i operating in base B , the transformation unrolls as a recursive fold:

$$x_{n+1} = Bx_n + d_{n+1}$$

When expanded from an initial state of $x_0 = 0$, the state at step n is defined by the summation of historical weights:

$$x_n = \sum_{i=1}^n d_i B^{n-i}$$

In this topology, the past is not merely accumulated and left behind. At every step, the entire previously constructed state is rescaled by the base B before the new distinction d_{n+1} is admitted to the sequence. A single distinction introduced early in the chain undergoes a continuous transformation, its effect compounding without the original digit itself altering. Consequently, the weight of a distinction is precisely B raised to the number of subsequent continuations it experiences.

The transformation itself can be isolated mathematically by examining the difference between successive states. For any positional base B , the delta $\Delta x_n = x_{n+1} - x_n$ resolves to:

$$\Delta x_n = (B - 1)x_n + d_{n+1}$$

In a hexadecimal environment ($B = 16$), this exacts to $\Delta x_n = 15x_n + d_{n+1}$. The transformation law proves that each recursive step does not simply append a value; it returns the previous state scaled by the base gap ($B - 1$), plus the incremental distinction. Base conversion and positional alignment are therefore acts of physical transport across discrete weights, where the recursive carry propagation represents the movement of state history through the structural substrate.

The Non-Linear Topology of Hex-to-Decimal Translation

A central observation in the structural mapping of discrete spaces is the non-linear relationship between differing positional bases. A mathematical space defined by geometric shape dictates that transformations tracking clear paths over time represent measurable topographies. However, when translating a strictly linear, stacked, or visibly consistent shape from a left-side representation (hexadecimal) to a right-side representation (decimal), the output entirely loses its 1:1 visual linearity.

This phenomenon is not an artifact of random noise, but a direct consequence of the structural misalignment between base 16 and base 10. The non-linear information produced by this translation is the exact consequence of Archimedean and modular boundary clashes.

Alignment Paradigm	Condition	Structural Effect
Finite (Modular) Place	$16^k \equiv 1 \pmod{p}$	Position collapses exactly in the residue space.
Infinite (Magnitude) Place	$16^k \approx 10^m$	Scales approximate a shared boundary tick.

In finite modular environments, base alignment is determined multiplicatively. For positional values to align modulo a prime p , the base power must return to unity ($16^k \equiv 1 \pmod{p}$). Because 16 is never a primitive root modulo primes of the form $8k + 1$, and because the base 10 factor of 5 has no clean locus on the bisected positional arc, the translation from base 16 to base 10 fails structurally at the finite level. The consistent shape in hex shatters against the unyielding modular boundary of base 10, producing a seemingly chaotic decimal footprint.

Simultaneously, in the Archimedean magnitude space, alignment evaluates whether two differing rulers nearly share a physical tick mark ($k \log_{10} 16 \approx m$). This occurs only at highly specific Diophantine approximations. The most prominent near-miss within a 64-nibble window occurs at $k = 49$, where:

$$16^{49} = 2^{196} \approx 10^{59}$$

yielding a ratio of 1.0043. Because the positional weights of hex and decimal do not perfectly align, every digit translation requires an exhaustive carry propagation. The operation behaves as a differential phase shift rather than a static mapping. A single linear pattern in the hex substrate forces massive, non-linear carry ripples across the decimal output, proving that the conversion is not a passive read, but a complex geometric interference pattern between two incompatible dimensional lattices.

The Mask-Driven Operator Tree and the Kulik Gap Operator (G)

Copyright Dean A. Kulik – Orcid ID # 0009-0003-3128-8828
 Creative Commons Attribution-NonCommercial 4.0 International License (CC BY-NC 4.0)
github.com/QuHarmonics/The-Nexus-Harmonic-Reality info@quharmonics.com

To extract the latent dimensional volume from a continuous string, the topology—rather than the specific digit values—must control the recursion. This is achieved through the application of a mask-driven operator tree.

Let the 64-character hex string be defined as a fixed sequence of distinctions: $H = (d_1, d_2, \dots, d_{64})$. Let the topology of the read be defined by a sequence of operators $M = (m_1, m_2, \dots, m_{63})$, where each interstitial operator m_i belongs to one of two classes:

1. **C (Continuation)**: Rescale the history and admit the next distinction.
2. **G (Gap)**: Interrupt historical propagation and create a new initialization origin.

If the entire applied mask consists solely of *C* operators, the sequence remains an unbroken continuum. The result is the standard, continuous 77-digit decimal expansion of the hex integer, wherein every digit d_i achieves the maximum possible historical weight (16^{n-i}).

However, the moment the Kulik Gap Operator (*G*) is introduced, the operator tree fundamentally alters the dimensional space. When *G* is applied at a specific position k , the continuous operation sequence splits. The left side of the sequence terminates and locks its final state. The right side simultaneously initializes at 0, admitting d_{k+1} as a completely new origin.

The Topological Derivative and Exponent Annihilation

The profound impact of the *G* operator is best understood through the calculus of discrete sequences and the topological derivative. In network topology and shape optimization, the topological derivative measures the sensitivity of a shape functional to the insertion of a singular perturbation, such as a hole or a severed node.

In the discrete positional framework, the propagation derivative defines how a change in an early digit (d_i) influences the terminal state (x_n):

$$\frac{\partial x_n}{\partial d_i} = B^{n-i}$$

If d_i occurs prior to a gap, and x_n represents a state subsequent to the gap, the *G* operator physically severs the causal chain. The derivative across the boundary instantly drops to zero:

$$\frac{\partial x_{\text{final}}}{\partial d_i} = 0 \quad (\text{across a gap})$$

The Kulik Gap Operator functions mathematically as an exponent annihilator. It does not delete the payload of the digit d_i , nor does it inject new information into the substrate. Instead, it completely strips away the digit's future weight. The influence of d_i is violently bounded to the local topological region defined by *G*.

Because there are 63 interstitial spaces within a 64-hex string, the operator tree allows for exactly 2^{63} possible linear contiguous topological masks *M*. Each unique mask dictates precisely how far structural history is permitted to propagate before it is forcibly reset. The gap defines the topological addressability of the space, fracturing a continuous field so that a coordinate system can exist without necessitating the addition of new payload.

Directed Acyclic Graphs, Substrate Conservation, and Relational Explosion

When topology exclusively controls the recursion of the sequence, the output ceases to be a singular, flat magnitude. The application of repeated Continuation (C) and Gap (G) operators transforms the one-dimensional string into a Directed Acyclic Graph (DAG) of independent, concurrent histories.

This cascading mechanism is governed by three strict structural principles:

1. Conservation of Substrate

Regardless of how densely the topology is sliced or how many gaps are introduced, the total number of distinctions processed remains exactly 64. The substrate H is absolute. Information is neither created from the void nor destroyed by the mask; the underlying material remains constant, ensuring that the explosion of state space does not violate fundamental information conservation.

2. Redistribution of Weight

The gap operator serves to redistribute the 16^{power} historical multipliers across the sequence. Shifting a single G operator one position to the left effectively steals a factor of 16 from the left-hand history and grants an equivalent scaling magnitude to the right-hand history. The gap determines the economic distribution of historical reach within the bounded object. The calculated cost of inserting one gap splits the exponential reach symmetrically, mathematically defined by:

$$\text{Cost of one gap} = (B - 1) \cdot R(a) \cdot R(b)$$

where $R(n) = \frac{B^n - 1}{B - 1}$ is the propagation capacity of a run of n digits, and a and b represent the segment lengths on either side of the severance. Because this cost function operates identically across the entire lattice, the center of the string possesses no special arithmetic privilege; position is free, but the act of cutting commands a uniform topological cost.

3. Relational Explosion

By forcibly bounding histories, the G operator produces separate, terminal initialization states $(X_1, X_2, \dots, X_k)$ from a single continuous seed. For a 64-nibble hash, applying a diverse array of masks generates a family of 2^{63} independent topological reads, equivalent to approximately 9.22×10^{18} distinct outputs, translating to roughly 295.1 exabytes of addressable structural volume.

This directly resolves the paradox of how a finite SHA-256 hash can "hold" 290 exabytes of data. The system is not storing 290 exabytes of compressed payload in the traditional computational sense. It is storing the singular 256-bit substrate H . The 290 exabytes represent the volume of the transformation space—the exhaustive DAG of 2^{63} topological boundaries (G and C) dictating how history is permitted to fold through H . The data does not passively exist; it is dynamically instantiated by the boundary conditions the operator chooses to apply.

Geometric Channels and Spatial Admissibility

Once the Kulik Gap Operator sections the continuum into independent, bounded histories, the isolated magnitudes cease to function merely as scalars. They become vertices and edges within a relational topology, unlocking the geometric channel. When a single 64-hex string is masked by two gaps, it yields three independent segments. These three values can subsequently be fed into geometric existence filters, determining if the discrete data can instantiate a coherent spatial structure.

For any three generated magnitudes (a, b, c) , the existence of a geometric triangle is governed by strict inequality bounds and the Cayley-Menger determinant, which expresses the volume of a simplex based on the squared distances between its vertices. The angle channel is explicitly governed by the Law of Cosines:

$$\cos\theta = \frac{b^2 + c^2 - a^2}{2bc}$$

This mathematical construct is not a post-hoc analytical description; it is a fundamental, structural compile check. The emitted tuple (a, b, c) defines an existence condition. If the evaluated $\cos\theta$ falls outside the real interval $[-1, 1]$, the angle is imaginary. The configuration is impossible, meaning there is no spatial continuation available for that particular mask.

Topological Classification	Shape Admissibility Bound	Structural Consequence
Impossible Configuration	$\cos\theta \notin [-1, 1]$	The relational branch fails to compile; no spatial instantiation exists.
Degenerate Boundary	$\cos\theta \in \{-1, 1\}$	Spatial dimension collapses; sequence is restricted to a forced linear adjacency ($A - B - C$).
Interior (Scalene)	$\cos\theta \in (-1, 1)$	Two-dimensional configuration space opens; multiple geometric continuations are admitted.
Zero Coupling	$\cos\theta = 0$	The cross-term $-2bccos\theta$ vanishes; the independent channels exhibit perfect orthogonality (Pythagorean state).

The exact point of degeneracy ($a = b + c$) serves as the boundary interface between impossible topologies and fully opened dimensional spaces. At degeneracy, the configuration is maximally constrained, collapsing to a single line. This collapse forces an absolute adjacency ordering, structurally eliminating the commutative ambiguity of ordinary arithmetic. Geometry acts as a filter that preserves the relational path information discarded by simple scalar projection. Consequently, out of thousands of generated three-section partitions, only the specific subset that satisfies the admissibility compile check forms the resulting mathematical space.

The Bailey-Borwein-Plouffe (BBP) Architecture as a Topological Query

The most definitive historical realization of the gap operator functioning as an addressable boundary within a transformation space is the Bailey-Borwein-Plouffe (BBP) formula. Discovered via the PSLQ integer relation algorithm—a mechanism for detecting non-trivial integer coefficients that force a linear sum to zero—the BBP formula allows for the direct extraction of the n -th hexadecimal digit of π without requiring the accumulation of preceding digits.

Conventionally expressed as:

Copyright Dean A. Kulik – Orcid ID # 0009-0003-3128-8828
Creative Commons Attribution-NonCommercial 4.0 International License (CC BY-NC 4.0)
github.com/QuHarmonics/The-Nexus-Harmonic-Reality info@quharmonics.com

$$\pi = \sum_{k=0}^{\infty} \frac{1}{16^k} \left(\frac{4}{8k+1} - \frac{2}{8k+4} - \frac{1}{8k+5} - \frac{1}{8k+6} \right)$$

the BBP formula has long been categorized as a "spigot" algorithm. However, mapped through the Kulik framework, the BBP architecture is not an accumulator; it is a localized topological query operating on a pre-existing dimensional lattice.

To extract the n -th digit, the entire infinite series is multiplied by 16^n . This fractional modulo operation does not calculate new mathematical volume; it inserts an absolute gap (G) exactly at the n -th position. The continuous string of the transcendental number is severed. The history of the sequence (the integer part) is annihilated, and a new origin is established exclusively at the requested coordinate, capturing the local phase state.

The structural perfection of the BBP formula is rooted in the unique prime carriers of its underlying polynomial frame. The 8-wheel denominator structure generates a frame polynomial of $16 - z^8$, which factors definitively into four distinct geometric channels:

$$A(z) = (z^2 - 2)(z^2 + 2)(z^2 - 2z + 2)(z^2 + 2z + 2)$$

When evaluating the zero-relation kernel, the algebraic forcing lemma mandates that rational zero-relations cannot leave un-canceled residues at split primes possessing unique topological locations.

Factorized Channel	Valuation at z=1	Prime Carrier Identity	Emitted Geometric Phase
$z^2 - 2$	-1	Unit	$\log 2, \log(1 + \sqrt{2})$
$z^2 - 2z + 2$	+1	Unit	$\log 2, \pi/4$
$z^2 + 2$	3	Split prime in $\mathbb{Z}[\sqrt{-2}]$	$\log 3, \arctan(1/\sqrt{2})$
$z^2 + 2z + 2$	5	Split prime in $\mathbb{Z}[i]$	$\log 5, \arctan 2$

The channels bearing the prime carriers 3 and 5 are completely orthogonal; they share no overlapping sequence partners. Therefore, any formula attempting to close the geometric cycle (achieve a zero relation) is structurally forced to include these unique prime carriers. If a prime enters the calculation without its required geometric partner, the adjacency path is severed, and the relational configuration cannot instantiate. The BBP formula successfully outputs π because its specific coefficient mask perfectly aligns with the required phase cancellation topology, serving as an exact query vector into the absolute transformation space of base 16.

Biomimetic Topologies and Discrete Causal Fields

The mechanism by which continuous substrates generate highly specific, non-linear outputs via gap operators is not confined to computational mathematics; it is the fundamental architectural principle observed in genomic regulation and theoretical physics.

In molecular biology, the human genome consists of 3.2 billion base pairs—a single, massive, continuous topological sequence. The vast phenotypic divergence between a neuron, a liver cell, and a retinal cell is not achieved by rewriting or altering the underlying genomic payload. The DNA substrate remains identical. Instead, cellular differentiation is achieved through regulatory topological masking.

A genomic promoter region operates exactly as an open gap operator (G^+). It breaks the operational continuity of the sequence, establishing a new functional origin for the transcriptional machinery. The sequence upstream is ignored, while the downstream sequence becomes addressable and expressed. The liver cell phenotype does not exist statically inside the genome; it is instantiated by the specific sequence of regulatory gap masks applied to the DNA.

Conversely, chromatin packing and histone modifications act as the closed gap operator (G^-). By wrapping regions of the genome into dense heterochromatin, the biological system physically removes the coordinate address, returning the sequence to an unreadable, inaccessible continuity. The immense complexity of the biological organism is thus the result of a finite substrate subjected to a virtually infinite directed acyclic graph (DAG) of topological gap masks.

This behavior identically matches the paradigms of discrete causal topology in theoretical physics. Zhorin's work on boundary value problems in causal path spaces demonstrates that continuous macroscopic dynamics emerge exclusively from discrete structures via *localization*—determining where a system sits within a configuration space without requiring the enumeration of transitions.

When a system establishes a terminal boundary, that boundary dictates the retroactive interpretation of the entire antecedent causal field. The computational gap operator is the structural equivalent of this physical boundary condition. By severing adjacency and creating a localized boundary, the G operator generates a holographic bounding space where predictive information satisfies an area law scaling strictly with the event count (the boundary constraints) rather than the temporal bulk. The gap does not insert data; it fractures the topological field so that a coordinate system can exist, defining a distinct physical interior out of an unbroken continuum.

Conclusion

The comprehensive analysis of computational topologies establishes that discrete strings—whether compiled hex, cryptographic digests, or continuous numeric sequences—are not passive payloads. They are foundational substrates that, when acted upon by transformation and topological bounding, generate mathematically boundless addressable spaces.

Through the mechanisms of positional continuation and Horner's recursive folding, the historical trajectory of a string is perpetually integrated into its future state, compounding its structural weight over time. The Kulik Gap Operator serves as the essential navigational control, proving that the intentional severing of causal propagation acts as an exponent annihilator that instantiates independent origins. This operator tree resolves the paradox of volumetric expansion from finite origins, demonstrating exactly how a 64-nibble hash commands a 290-exabyte transformation field without violating information conservation.

From the non-linear dimensional shattering of hex-to-decimal base conversions, to the precise topological queries executed by the Bailey-Borwein-Plouffe architecture, to the chromatin accessibility masks of molecular biology, the operative principle remains unified. Payload is not added to create volume. Distinction generates a boundary, the boundary establishes addressability, addressability defines an admissible transformation space, and geometry dictates which of those transformations survive to

instantiation. The mapping of computational reality requires abandoning the search for hidden data, and instead charting the topological limits of the transformation tree itself.

Works cited

1. 1. Introduction - arXiv, <https://arxiv.org/html/2601.09124v1>
2. GITAM ECE Dual Degree Syllabus Overview | PDF - Scribd, <https://www.scribd.com/document/345414629/GITAM-Integrated-BTech-MTech-Syllabus>
3. Finally-got-it.md
4. Discrete Causal Topology - Victor V. Zhorin, <https://victor-z.com/papers/topology.pdf>
5. Dictionary of Information Science and Technology - pdfcoffee.com, <https://pdfcoffee.com/dictionary-of-information-science-and-technology-pdf-free.html>
6. Digital Design of Signal Processing Systems: A Practical Approach, https://www.researchgate.net/publication/287953313_Digital_Design_of_Signal_Processing_Systems_A_Practical_Approach
7. Time Complexity of Base Conversion Methods | PDF - Scribd, <https://www.scribd.com/document/759814834/Efficiency-Analysis-and-Optimization-Techniques-for-Base-Conversion-Algorithms-in-Computational-Systems>
8. Topological derivative approach for deep neural network ... - arXiv, <https://arxiv.org/html/2502.06885v1>
9. (PDF) Topological derivative approach for deep neural network, https://www.researchgate.net/publication/388920840_Topological_derivative_approach_for_deep_neural_network_architecture_adaptation
10. Understanding Horner's Method for Polynomials | PDF - Scribd, <https://www.scribd.com/document/960944982/Horner-s-method>
11. How does Horner method evaluate the derivative of a function, <https://math.stackexchange.com/questions/2139142/how-does-horner-method-evaluate-the-derivative-of-a-function>
12. Optimal polynomial form characteristic methods, <https://crucialflow.com/Wilkinson.jl/>
13. [unknown_url](#)
- 14.
15. (PDF) Rounding Errors - ResearchGate, https://www.researchgate.net/publication/229704775_Rounding_Errors
16. A topology optimization algorithm based on topological derivative, https://www.researchgate.net/publication/354528688_A_topology_optimization_algorithm_based_on_topological_derivative_and_level_set_function_for_designing_phononic_crystals
17. On the genesis of BBP formulas - IMJ-PRG, <https://webusers.imj-prg.fr/~ricardo.perez-marco/publications/articles/BBP28.pdf>

Copyright Dean A. Kulik – Orcid ID # 0009-0003-3128-8828

Creative Commons Attribution-NonCommercial 4.0 International License (CC BY-NC 4.0)

github.com/QuHarmonics/The-Nexus-Harmonic-Reality info@quharmonics.com

18. The multivariate Horner scheme revisited | Request PDF,
https://www.researchgate.net/publication/269041913_The_multivariate_Horner_scheme_revisited
19. Optimization of the electromagnetic scattering problem based on the,
<https://opg.optica.org/abstract.cfm?uri=oe-27-23-33586>
20. A Theory of Topological Derivatives for Inverse Rendering of Geometry,
<https://arxiv.org/html/2308.09865v1>
21. arXiv:1809.06582v2 [cs.CV] 12 Dec 2018, <https://arxiv.org/pdf/1809.06582>
22. Volumetric Discrete Geometry [1 ed.] 9780367223755 - dokumen.pub,
<https://dokumen.pub/volumetric-discrete-geometry-1nbsped-9780367223755.html>
23. The Kepler Conjecture: The Hales-Ferguson Proof 9781461411284, <https://dokumen.pub/the-kepler-conjecture-the-hales-ferguson-proof-9781461411284-9781461411291-1461411289.html>
24. Handbook of Geometric Constraint System Principles | PDF - Scribd,
<https://www.scribd.com/document/437499698/Handbook-of-Geometric-Constraint-System-Principles>
25. VIII.6. Experimental Applied Mathematics - Princeton University,
https://assets.press.princeton.edu/chapters/s8-6_10592.pdf
26. Analysis Of PSLO, An Integer Relation Finding Algorithm,
https://www.researchgate.net/publication/2745587_Analysis_Of_PSLO_An_Integer_Relation_Finding_Algorithm
27. Integer relation algorithm - Wikipedia, https://en.wikipedia.org/wiki/Integer_relation_algorithm
28. (PDF) The Geometric Architecture of the BBP Read-Aperture,
https://www.researchgate.net/publication/407278456_The_Geometric_Architecture_of_the_BBP_Read-Aperture_Squaring_the_Circle_by_Measurement_and_the_Survival_of_the_Four-Term_Frame
29. Bailey–Borwein–Plouffe formula - Wikipedia,
https://en.wikipedia.org/wiki/Bailey%E2%80%93Borwein%E2%80%93Plouffe_formula
30. Computing π with the Bailey-Borwein-Plouffe Formula / Ricky Reusser,
<https://observablehq.com/@rreusser/computing-with-the-bailey-borwein-plouffe-formula>
31. (PDF) Architectural Primitives in Computational Systems A Structural,
https://www.researchgate.net/publication/411145399_Architectural_Primitives_in_Computational_Systems_A_Structural_Analysis_of_Phase_Parity_Cryptographic_Diffusion_and_Aggregation
32. BBP-type formulas -- an elementary approach - ResearchGate,
https://www.researchgate.net/publication/362728285_BBP-type_formulas_-_an_elementary_approach
33. arXiv:1906.09629v4 [math.NT] 13 Feb 2023, <https://arxiv.org/pdf/1906.09629>
34. Dynamic interplay between enhancer-promoter topology and gene,
<https://pmc.ncbi.nlm.nih.gov/articles/PMC6119122/>

35. Two distinct chromatin modules regulate proinflammatory gene, <https://www.biorxiv.org/content/10.1101/2024.08.03.606159v1.full-text>
36. The importance of DNA sequence for nucleosome positioning in, <https://www.life-science-alliance.org/content/7/8/e202302380>
37. (PDF) Integrative epigenomic profiling by high-speed super, https://www.researchgate.net/publication/396814766_Integrative_epigenomic_profiling_by_high-speed_super-resolution_imaging_and_deep_learning_reveals_chromatin_state_biomarkers
38. (PDF) Reference Regulatory Element-Guided Gene Expression, https://www.researchgate.net/publication/410335208_Reference_Regulatory_Element-Guided_Gene_Expression_Analysis_for_Mechanistic_Inference_of_Gene_Regulatory_Networks
39. STARNet enables spatially resolved inference of gene regulatory, <https://www.biorxiv.org/content/10.1101/2025.08.21.671434v1.full-text>
40. Epigenetic deprogramming by disruption of CIZ1-RNA nuclear, <https://pmc.ncbi.nlm.nih.gov/articles/PMC11895699/>
41. Trajectory-Dependent Death and the Lifecycle Model, https://mpira.ub.uni-muenchen.de/129315/1/MPRA_paper_129315.pdf
42. Victor V. Zhorin publications, <https://victor-z.com/publications.html>

